

UNDERSTANDING AND CONCEPTUALIZING INTERACTION

資工碩一 俞尚毅

資工碩一 江秉翰

電資大四 蔡樂津

人社大四 虞佩君

Understanding the Problem Space and Conceptualizing Interaction

01

WHAT(DEFINITION)

Understand and conceptualizing what is currently the user experience/product and how this is going to be improved or changed.

1. What do you want to create?
2. What are your assumptions?
3. Will it achieve what you hope?
4. ...

02

WHY(PROS & CONS)

1. Time-consuming and resulting in disagreements
2. Much less chance of incorrect assumptions and unsupported claims that later turn out to be unusable or unwanted
3. Enables more options and possibilities to be considered

03

HOW(GUIDELINE)

1. Identifying usability and user experience goals
2. Make explicit underlying **assumptions** and **claims**

03

INCORRECT ASSUMPTIONS

Taking something for granted when it needs further investigation

UNSUPPORTED CLAIMS

Stating something to be true when it is still open to question

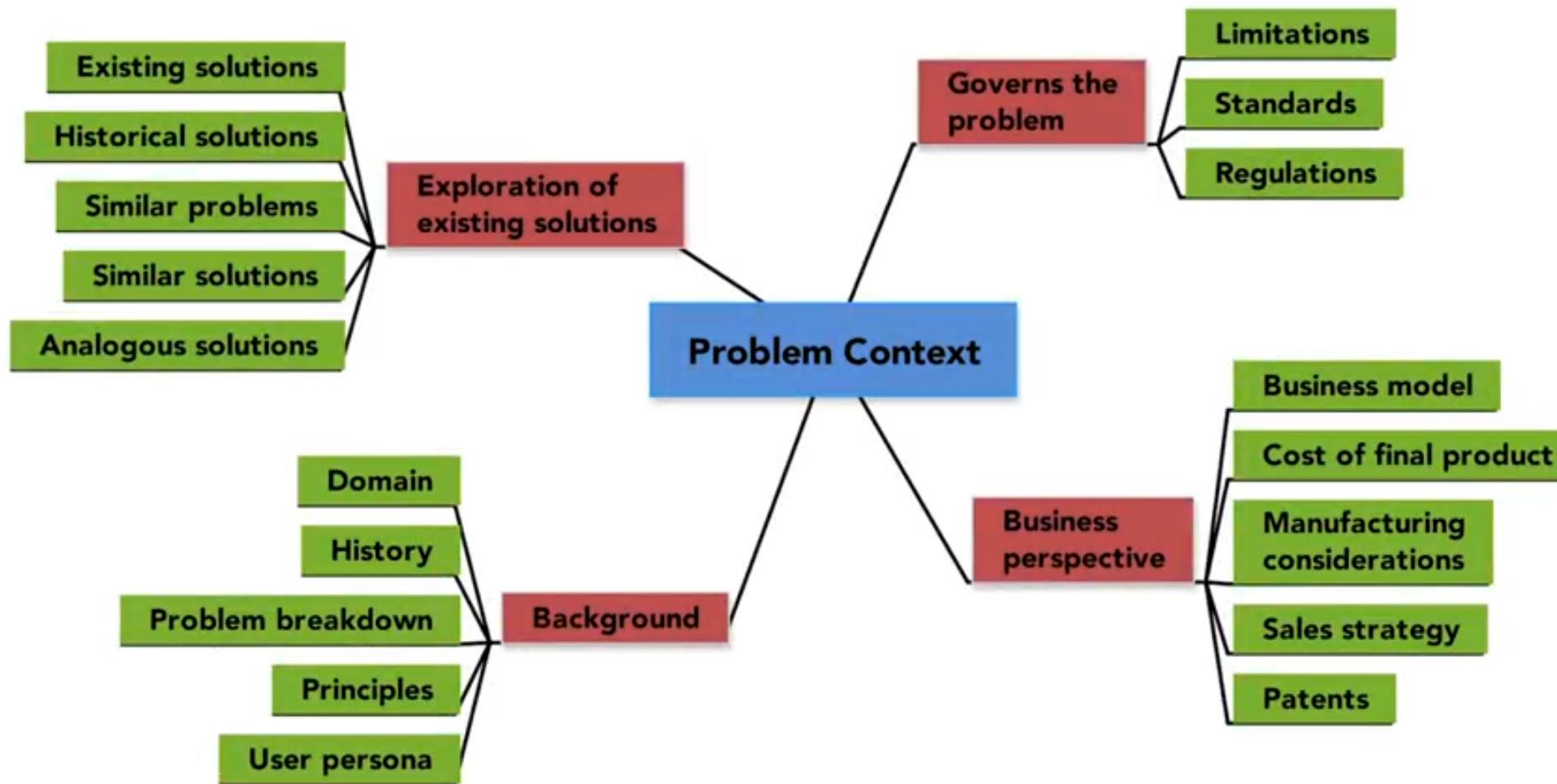
TV system



03

HOW(FRAMEWORK)

1. Are there problems with the product or user experience? If so, what are they?
2. Why do you think they are problems?
3. How do you think your proposed design ideas might overcome these?
4. How do you think your proposed design ideas might support, change or extend current ways of doing thing?



04

FROM **PROBLEM SPACE** TO **DESIGN SPACE**

1. Understand the problem space
2. Develop a conceptual model

BENEFITS

Orientation, Open-mindedness, Common ground

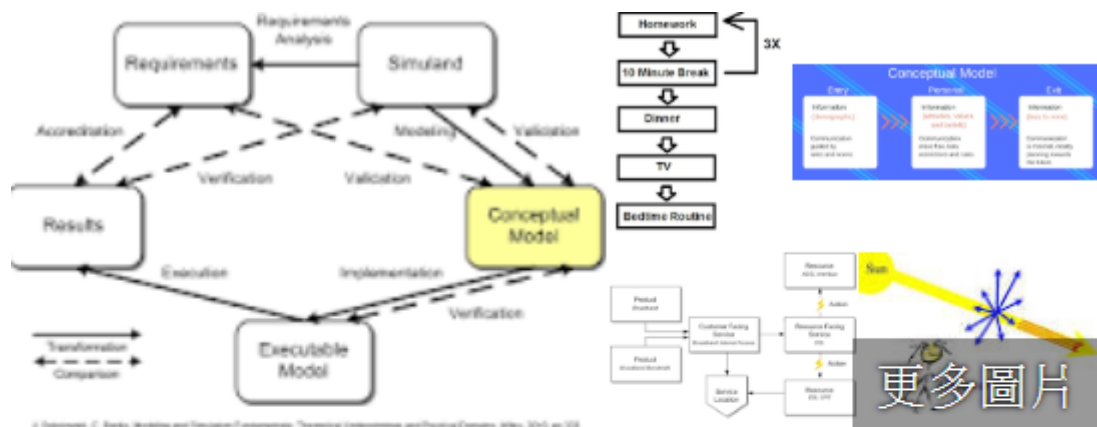
Conceptual Models

Definition of Conceptual Models

- In textbook
 - “A high-level description of how system is organized and operates”
 - Provide a working strategy and a framework of general concepts and their interrelations
 - Related to **user experience**
 - 一種概念模型 (介面), 能讓使用者知道如何使用產品 (出於本能, 認知)

Definition of Conceptual Models

- From wikipedia



概念模型

概念模型在電腦人機互動領域中，概念模型指的是關於某種系統一系列在構想、概念上的描述，敘述其如何作用，能讓使用者瞭解此系統被設計師預設之使用方式。

[維基百科](#)

Definition of Conceptual Models

- From website (<https://airbrake.io/blog/sdlc/conceptual-model>)
 - Conceptual modeling is used as a way to describe physical or social aspects of the world in an **abstract way**
 - Should fulfill some objectives
 - Enhance understanding of the representative system.
 - Promote efficient conveyance (表達) of system details between team members.

Conceptual Models

- Best conceptual models appear obvious
 - 簡單易懂
- **Metaphors** (隱喻) is one central component of conceptual model

Interface Metaphors

Interface Metaphors

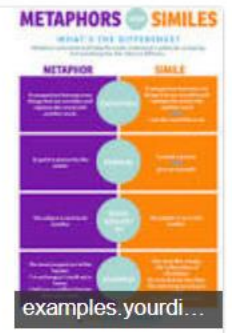
- Interface metaphors are intended to provide familiar entities that enable people to readily understand the underlying conceptual model and know what to do at an interface



全部 圖片 新聞 影片 地圖 更多

約有 63,900,000 項結果 (搜尋時間：0.55 秒)

Metaphor Examples. ... Simply put, a **metaphor** is a figure of speech containing an implied comparison. With **metaphors**, words or phrases that are ordinarily applied to one thing are applied to something you wouldn't necessarily pair it with. Here's a **metaphor example**: "The curtain of night fell upon us."



[Metaphor Examples - Examples on YourDictionary](https://examples.yourdictionary.com/metaphor-examples.html)
examples.yourdictionary.com/metaphor-examples.html

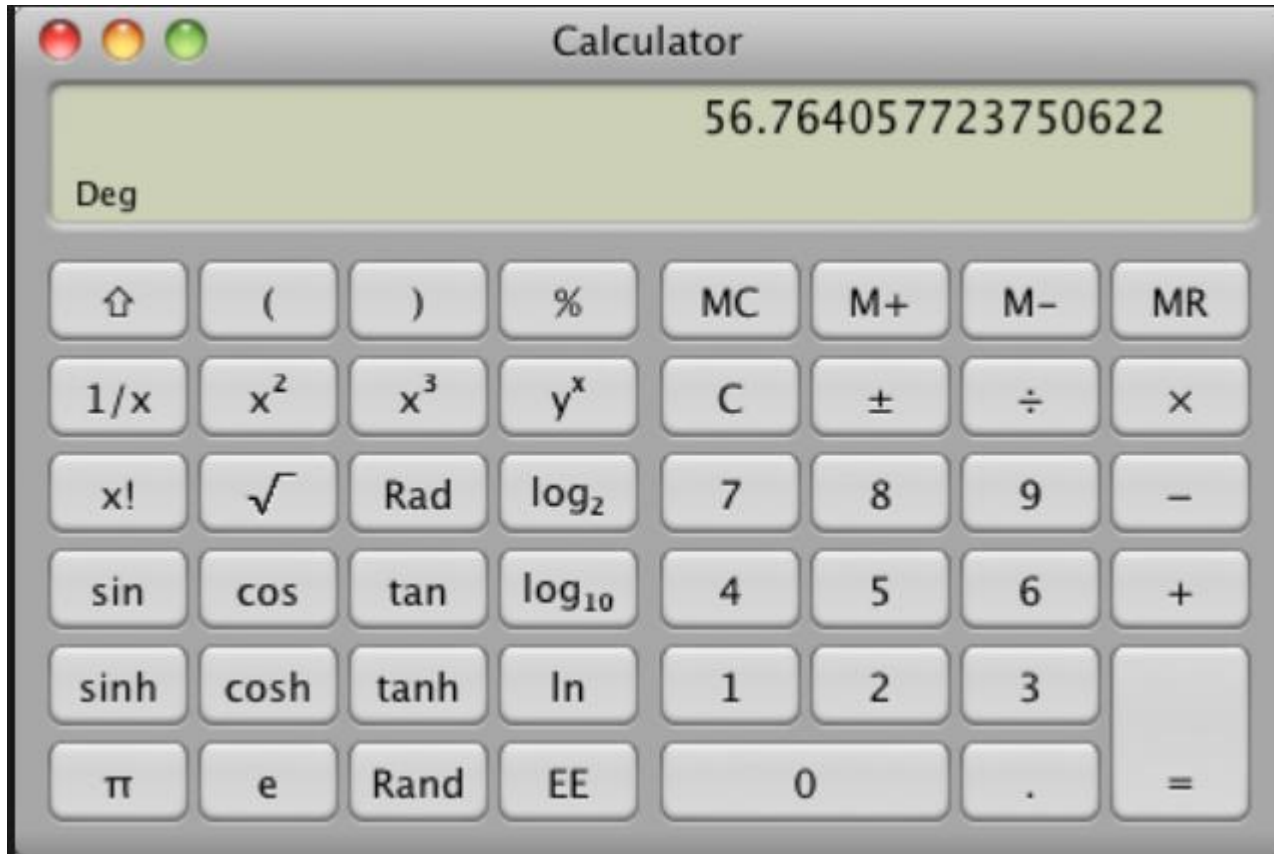
進一步瞭解這項結果 | 意見回饋

[Metaphor Examples for Kids - Examples on YourDictionary](https://examples.yourdictionary.com/metaphor-examples-for-kids.html)
[examples.yourdictionary.com](https://examples.yourdictionary.com/metaphor-examples-for-kids.html) > ... > [Metaphor Examples for Kids](https://examples.yourdictionary.com/metaphor-examples-for-kids.html) ▾ 翻譯這個網頁

Metaphor Examples for Kids. A **metaphor** is a figure of speech that is used to make a comparison between two things that aren't alike but do have something in common. Unlike a simile, where two things are compared directly using like or as, a **metaphor's** comparison is more indirect by stating something is something else.

Interface Metaphors

- Bad example from textbook



Interaction Types

What is Interaction Types?

- Essentially, these are the **ways** a person interact with a product or an application.
- It helps to conceptualize the design space.

Interaction Types

1. Instructing
 - where users issue instructions to a system
2. Conversing
 - where users have a dialog with a system
3. Manipulating
 - where users interact with objects in a virtual or physical space by manipulating them
4. Exploring
 - where users move through a virtual or physical space

Interaction Types - Instructing

- Users instruct a system and tell it what to do directly.
- Main benefit is that instructing supports quick and efficient interaction.

Interaction Types - Instructing



Vending Machine

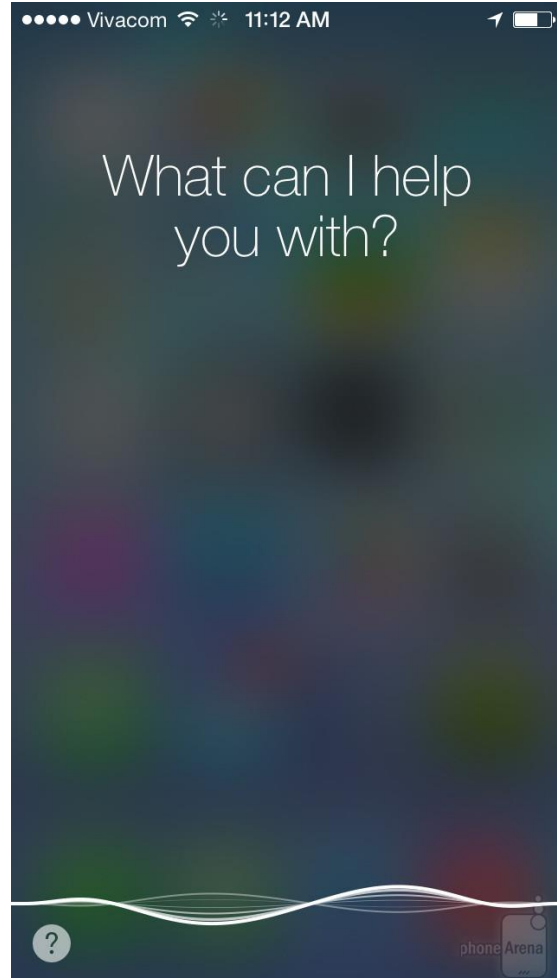
```
C:\Windows\system32\cmd.exe
C:\Users\limited\Desktop>es -size -dm -sizecolor 13 -dmcolor 11 -sort dm -n 10
 78,440 26/10/2016 20:02 C:\Users\limited\Desktop\es.exe
2,023,424 26/10/2016 13:57 C:\Users\limited\Desktop\Everything.exe
 32,768 21/10/2016 16:37 C:\Boot\BCD
1,048,576 21/10/2016 16:37 C:\Users\limited\NTUSER.DAT
 53,767 21/10/2016 16:37 C:\Users\limited\AppData\Local\IconCache.db
268,435,456 21/10/2016 16:08 C:\swapfile.sys
738,197,504 21/10/2016 16:08 C:\pagefile.sys
387,332 19/10/2016 19:31 C:\Users\limited\Desktop\Everything-Debug.txt
 181 19/10/2016 18:47 C:\Users\limited\Desktop\Everything.ini
7,328,537 14/10/2016 23:25 C:\dev\Everything.db
C:\Users\limited\Desktop>_
```

Command Line Interface

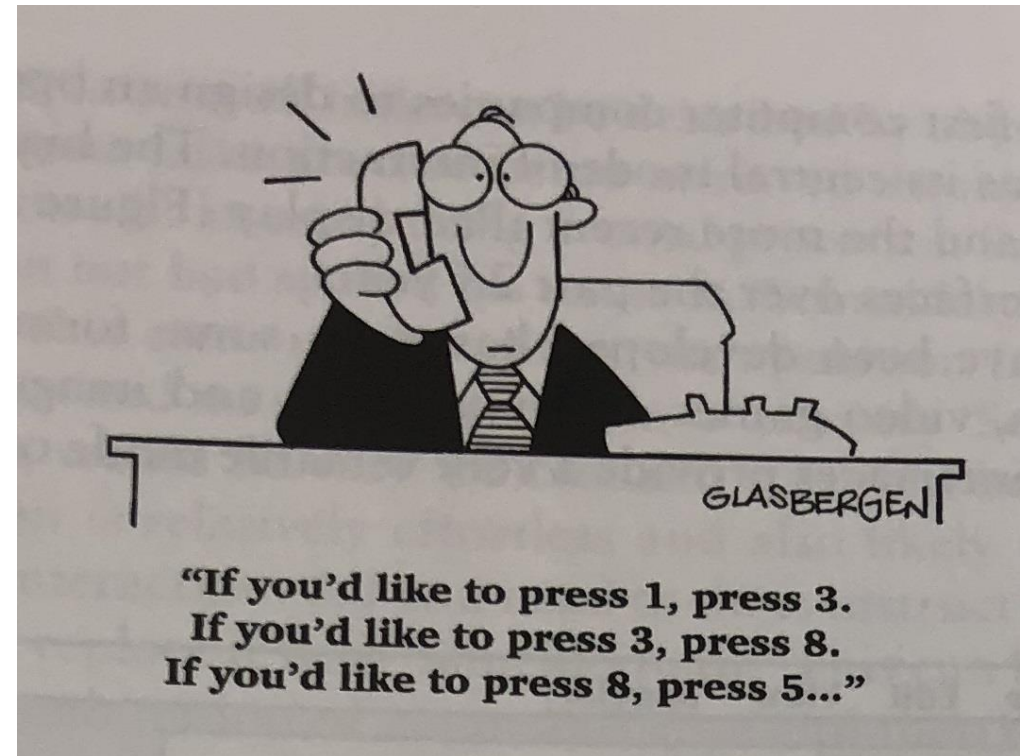
Interaction Types - Conversing

- Underlying model of having a conversation with another human.

Interaction Types - Conversing



Siri

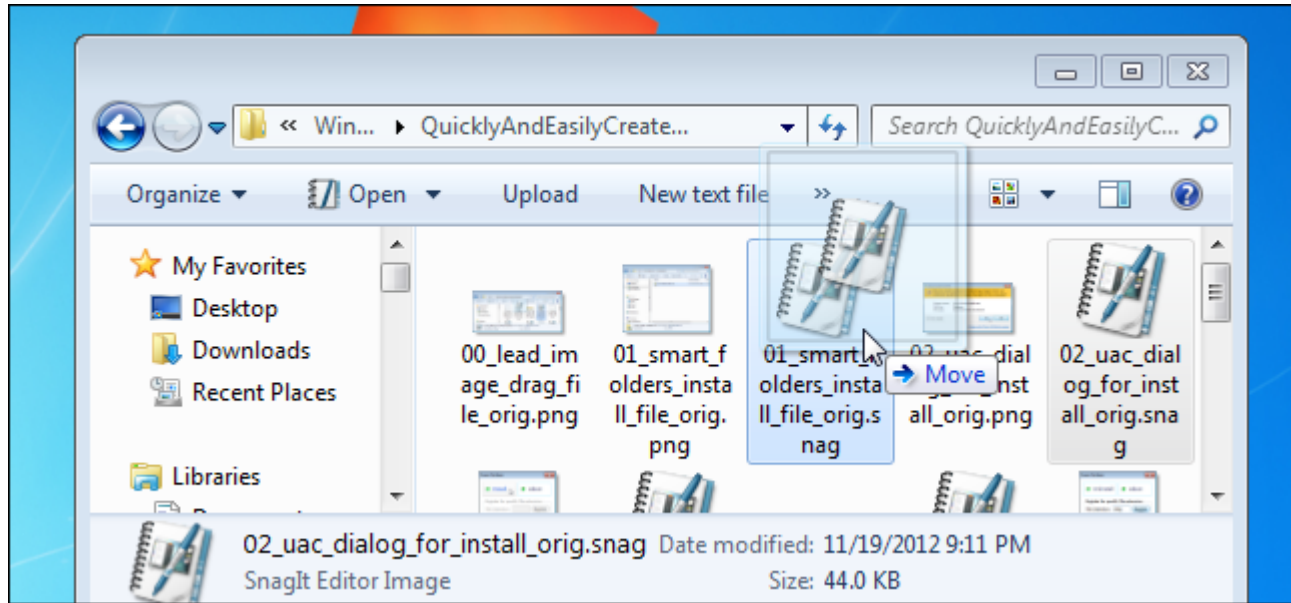


Automated Phone-based System

Interaction Types - Manipulating

- Involving dragging, selecting, opening, closing and zooming actions on virtual objects.
- Exploiting users' knowledge of how they move and manipulate in the physical world.

Interaction Types - Manipulating



Dragging files on Windows

Interaction Types - Exploring

- Involving users moving through virtual or physical environment.
- Physical environments with embedded sensor technologies.

Interaction Types - Exploring



Digital Museum Guide



Virtual World

Other Tools

Paradigms

- A general approach that has been adopted by a community of researchers and designers for carrying out their work.
- Sharing assumptions, concepts, values, and practices.
- Kuhn認為Paradigm是“特定的科學共同體從事某一類科學活動所必須遵循的公認的模式”。（引自MBA智庫百科）

Theories

- Explanation of a phenomenon.
- Can help identify factors.

Models

- A simplification of some aspect of HCI .
- Intended to make it easier for designers to predict and evaluate alternative designs.

Frameworks

- A set of interrelated concepts and/or specific questions.
- Intended to inform a particular domain area, online communities, or an analytic method.

Thanks for your attention!